

2.3 Expectation of quadratic forms

Sums of squares are quadratic forms in the response. The sample variance is $(n-1)^{-1}\mathbf{Y}^\top(\mathbf{I} - n^{-1}\mathbf{J})\mathbf{Y}$. The residual sum of squares of a regression is $\mathbf{Y}^\top(\mathbf{I} - \mathbf{M})\mathbf{Y}$, where $\mathbf{M}$ is the projection onto the column space of $\mathbf{X}$ (Chapter 6). Test statistics are ratios of such forms. Their expectations follow from a short trace argument that needs no distributional assumption.

2.3.1 The basic formula

Theorem 2.3.1 — Mean of a quadratic form

Let $\mathbf{Y}$ be a $p \times 1$ random vector with finite second moments, $E(\mathbf{Y}) = \boldsymbol{\mu}$ and $\text{Cov}(\mathbf{Y}) = \boldsymbol{\Sigma}$, and let $\mathbf{A}$ be a constant $p \times p$ matrix. Then

$$E(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = \text{tr}(\mathbf{A} \boldsymbol{\Sigma}) + \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}.$$

Proof. A 1×1 matrix equals its trace, and $\text{tr}(\mathbf{a}\mathbf{b}^\top) = \mathbf{b}^\top \mathbf{a}$ (Theorem 1.6.1), so $\mathbf{Y}^\top \mathbf{A} \mathbf{Y} = \text{tr}(\mathbf{A} \mathbf{Y} \mathbf{Y}^\top)$. By Proposition 2.1.1 and Proposition 2.1.2(b),

$$E(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = \text{tr}(\mathbf{A} E(\mathbf{Y} \mathbf{Y}^\top)) = \text{tr}(\mathbf{A}(\boldsymbol{\Sigma} + \boldsymbol{\mu} \boldsymbol{\mu}^\top)) = \text{tr}(\mathbf{A} \boldsymbol{\Sigma}) + \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}.$$

■

The formula has two parts. The trace term is what the form picks up from random variation. The second term is the value of the form at the mean, which is what it would be without any randomness. The theorem does not require independence, normality or symmetry of $\mathbf{A}$. Since $\mathbf{Y}^\top \mathbf{A} \mathbf{Y} = \mathbf{Y}^\top \mathbf{A}_s \mathbf{Y}$ with $\mathbf{A}_s = \frac{1}{2}(\mathbf{A} + \mathbf{A}^\top)$, and $\text{tr}(\mathbf{A} \boldsymbol{\Sigma}) = \text{tr}(\mathbf{A}_s \boldsymbol{\Sigma})$ when $\boldsymbol{\Sigma}$ is symmetric, we may assume $\mathbf{A}$ symmetric whenever that is convenient.

Corollary 2.3.2

Under the conditions of Theorem 2.3.1:

- (a) $E[(\mathbf{Y} - \mathbf{b})^\top \mathbf{A} (\mathbf{Y} - \mathbf{b})] = \text{tr}(\mathbf{A} \boldsymbol{\Sigma}) + (\boldsymbol{\mu} - \mathbf{b})^\top \mathbf{A} (\boldsymbol{\mu} - \mathbf{b})$ for any constant $\mathbf{b}$;
- (b) if $\boldsymbol{\Sigma} = \sigma^2 \mathbf{I}$, then $E(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = \sigma^2 \text{tr}(\mathbf{A}) + \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}$;
- (c) if $\boldsymbol{\Sigma}$ is positive definite, the squared Mahalanobis distance Eq. 2.2.2 has $E \Delta^2(\mathbf{Y}) = p$;
- (d) for random vectors $\mathbf{U}$ ($p \times 1$) and $\mathbf{V}$ ($q \times 1$) with finite second moments and a constant $p \times q$ matrix $\mathbf{A}$, $E(\mathbf{U}^\top \mathbf{A} \mathbf{V}) = \text{tr}(\mathbf{A} \text{Cov}(\mathbf{V}, \mathbf{U})) + E(\mathbf{U})^\top \mathbf{A} E(\mathbf{V})$.

Proof.

- (a) applies the theorem to $\mathbf{Y} - \mathbf{b}$, which has covariance matrix $\boldsymbol{\Sigma}$. (b) is immediate.
- (c) is (a) with $\mathbf{b} = \boldsymbol{\mu}$ and $\mathbf{A} = \boldsymbol{\Sigma}^{-1}$, so the trace is $\text{tr}(\mathbf{I}_p) = p$. For (d), $\mathbf{U}^\top \mathbf{A} \mathbf{V} = \text{tr}(\mathbf{A} \mathbf{V} \mathbf{U}^\top)$ and $E(\mathbf{V} \mathbf{U}^\top) = \text{Cov}(\mathbf{V}, \mathbf{U}) + E(\mathbf{V}) E(\mathbf{U})^\top$.

Example. (Sample variance with unequal means and variances)

Let $Y_1, \dots, Y_n$ be uncorrelated with means μ_i and variances σ_i^2 , and let $S^2 = (n-1)^{-1} \mathbf{Y}^\top \mathbf{C} \mathbf{Y}$ with $\mathbf{C} = \mathbf{I} - n^{-1} \mathbf{J}$. Here $\mathbf{\Sigma} = \text{diag}(\sigma_1^2, \dots, \sigma_n^2)$, so $\text{tr}(\mathbf{C}\mathbf{\Sigma}) = (1 - 1/n) \sum_i \sigma_i^2$, and $\boldsymbol{\mu}^\top \mathbf{C} \boldsymbol{\mu} = \sum_i (\mu_i - \bar{\mu})^2$. Therefore

$$\mathbb{E}(S^2) = \overline{\sigma^2} + \frac{1}{n-1} \sum_{i=1}^n (\mu_i - \bar{\mu})^2, \quad \overline{\sigma^2} = \frac{1}{n} \sum_{i=1}^n \sigma_i^2.$$

With a common mean, S^2 is unbiased for the *average* variance, even when the variances differ. Differences in the means inflate it. That is exactly why a residual mean square estimates σ^2 only when the mean structure is right.

Example. (Residual sum of squares under a wrong mean)

Let $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n$ and $\mathbb{E}(\mathbf{Y}) = \boldsymbol{\theta}$, where $\boldsymbol{\theta}$ need not lie in $\mathcal{C}(\mathbf{X})$. The residual sum of squares is $\mathbf{Y}^\top (\mathbf{I} - \mathbf{M}) \mathbf{Y}$, where $\mathbf{M}$ is symmetric and idempotent of rank $r = \text{rank}(\mathbf{X})$, so $\text{tr}(\mathbf{I} - \mathbf{M}) = n - r$ (Theorem 1.9.2). By [Corollary 2.3.2\(b\)](#),

$$\mathbb{E}(\text{RSS}) = \sigma^2(n - r) + \|(\mathbf{I} - \mathbf{M})\boldsymbol{\theta}\|^2. \quad (2.3.1)$$

If the model is right, $\boldsymbol{\theta} \in \mathcal{C}(\mathbf{X})$, the second term vanishes and $\text{RSS}/(n - r)$ is unbiased for σ^2 . If the model is wrong, the estimate is biased upwards by the squared distance from the true mean to the model space, divided by $n - r$. For instance, fit a straight line at $x = 1, \dots, 6$ when the true mean is the parabola $\theta_i = \frac{1}{4}(x_i - 3.5)^2$ and $\sigma^2 = 1$. Then $\sigma^2(n - r) = 4$, $\|(\mathbf{I} - \mathbf{M})\boldsymbol{\theta}\|^2 = 2.333$, and $\mathbb{E}(\text{RSS}) = 6.333$. Chapter 19 studies such departures systematically.

The theorem uses only two moments, so it holds equally for skewed and heavy-tailed data. The listing checks it for a correlated vector $\mathbf{Y} = \boldsymbol{\mu} + \mathbf{B}\mathbf{U}$, where $\mathbf{U}$ has independent, centred exponential components, which are strongly skewed. With the matrices in the listing, $\text{tr}(\mathbf{A}\mathbf{\Sigma}) = 6.9$ and $\boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu} = 17.0$, so $\mathbb{E}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = 23.9$. The average over 10^6 draws is 23.940, with Monte Carlo standard error 0.018. The script also verifies the formula exactly, by enumerating a discrete distribution with eight atoms.

```
import itertools
import numpy as np
rng = np.random.default_rng(20260917)

mu = np.array([1.0, -1.0, 2.0])
B = np.array([[1.0, 0.0, 0.0],
              [0.5, 1.0, 0.0],
              [-0.5, 0.8, 0.6]])
Sigma = B @ B.T
A = np.array([[2.0, 1.0, 0.0],
              [1.0, 1.0, -1.0],
              [0.0, -1.0, 3.0]])
```

Cov(mu + B U) when Cov(U) = I

```

theory = np.trace(A @ Sigma) + mu @ A @ mu          # Theorem: no normality needed

N = 1_000_000
U = rng.exponential(size=(N, 3)) - 1.0             # skewed, mean 0, variance 1
Y = mu + U @ B.T
Q = np.einsum("ij,jk,ik->i", Y, A, Y)              # Y_i' A Y_i for every draw
print(f"theory {theory:.4f}    simulation {Q.mean():.4f}  {Q.std() / np.sqrt(N):.4f}"
      )

```

[Run this code online](#)

2.3.2 The variance needs fourth moments

The mean of $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$ involves only $E(Y_i Y_j)$. Its second moment involves $E(Y_i Y_j Y_k Y_l)$, so no formula for $\text{Var}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y})$ can be written in terms of $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ alone. Two random vectors with the same mean and covariance matrix can give the same quadratic form very different variances. The next theorem handles independent components, the case that matters for residual sums of squares under the classical error assumptions. Chapter 4 gives the normal case with a general $\boldsymbol{\Sigma}$ (Theorem 4.2.3).

Theorem 2.3.3 — Variance of a quadratic form

Let $Y_1, \dots, Y_n$ be independent with means $\theta_1, \dots, \theta_n$ and common central moments $\sigma^2 = E(Y_i - \theta_i)^2$, $\mu_3 = E(Y_i - \theta_i)^3$ and $\mu_4 = E(Y_i - \theta_i)^4 < \infty$. Let $\mathbf{A}$ be a constant symmetric $n \times n$ matrix, and let $\mathbf{a} = (a_{11}, \dots, a_{nn})^\top$ be its diagonal. Then

$$\text{Var}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = (\mu_4 - 3\sigma^4) \mathbf{a}^\top \mathbf{a} + 2\sigma^4 \text{tr}(\mathbf{A}^2) + 4\sigma^2 \boldsymbol{\theta}^\top \mathbf{A}^2 \boldsymbol{\theta} + 4\mu_3 \boldsymbol{\theta}^\top \mathbf{A} \mathbf{a}.$$

Proof. Write $\mathbf{Y} = \boldsymbol{\theta} + \boldsymbol{\varepsilon}$, where the ε_i are independent with mean zero. Then

$$\mathbf{Y}^\top \mathbf{A} \mathbf{Y} = q + \ell + \boldsymbol{\theta}^\top \mathbf{A} \boldsymbol{\theta}, \quad q = \boldsymbol{\varepsilon}^\top \mathbf{A} \boldsymbol{\varepsilon}, \quad \ell = 2\mathbf{b}^\top \boldsymbol{\varepsilon}, \quad \mathbf{b} = \mathbf{A} \boldsymbol{\theta},$$

so $\text{Var}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = \text{Var}(q) + \text{Var}(\ell) + 2\text{Cov}(q, \ell)$. We compute the three terms.

The linear term. $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$, so $\text{Var}(\ell) = 4\sigma^2 \mathbf{b}^\top \mathbf{b} = 4\sigma^2 \boldsymbol{\theta}^\top \mathbf{A}^2 \boldsymbol{\theta}$.

The cross term. Since $E(\ell) = 0$, $\text{Cov}(q, \ell) = E(q\ell) = 2 \sum_{i,j,k} a_{ij} b_k E(\varepsilon_i \varepsilon_j \varepsilon_k)$. By independence and zero means, $E(\varepsilon_i \varepsilon_j \varepsilon_k)$ vanishes unless $i = j = k$, when it equals μ_3 . So $\text{Cov}(q, \ell) = 2\mu_3 \sum_i a_{ii} b_i = 2\mu_3 \boldsymbol{\theta}^\top \mathbf{A} \mathbf{a}$.

The quadratic term. $E(q^2) = \sum_{i,j,k,l} a_{ij} a_{kl} E(\varepsilon_i \varepsilon_j \varepsilon_k \varepsilon_l)$. The fourth-order moment is μ_4 if all four indices agree. It is σ^4 if the indices form two distinct pairs, which happens in three patterns: $i = j \neq k = l$, $i = k \neq j = l$ and $i = l \neq j = k$. Otherwise some index occurs exactly once, and the moment is zero. Using the symmetry of $\mathbf{A}$,

$$E(q^2) = \mu_4 \sum_i a_{ii}^2 + \sigma^4 \left[\sum_{i \neq k} a_{ii} a_{kk} + 2 \sum_{i \neq j} a_{ij}^2 \right] = \mu_4 \mathbf{a}^\top \mathbf{a} + \sigma^4 [(\text{tr } \mathbf{A})^2 - \mathbf{a}^\top \mathbf{a} + 2 \text{tr}(\mathbf{A}^2) - 2\mathbf{a}^\top \mathbf{a}],$$

since $\sum_{i,j} a_{ij}^2 = \text{tr}(\mathbf{A}^2)$. By Theorem 2.3.1, $E(q) = \sigma^2 \text{tr } \mathbf{A}$, and subtracting its square

gives $\text{Var}(q) = (\mu_4 - 3\sigma^4)\mathbf{a}^\top \mathbf{a} + 2\sigma^4 \text{tr}(\mathbf{A}^2)$. Adding the three terms proves the formula. ■

For normal errors, $\mu_3 = 0$ and $\mu_4 = 3\sigma^4$, and the formula collapses to $2\sigma^4 \text{tr}(\mathbf{A}^2) + 4\sigma^2 \boldsymbol{\theta}^\top \mathbf{A}^2 \boldsymbol{\theta}$. Both extra terms describe how non-normality leaks into second-order behaviour. The kurtosis term depends on $\mathbf{A}$ only through its diagonal. The skewness term needs both a skewed error distribution and a mean that the form does not annihilate. For the residual sum of squares of a correct model, $\mathbf{A}\boldsymbol{\theta} = (\mathbf{I} - \mathbf{M})\boldsymbol{\theta} = \mathbf{0}$, so the skewness term disappears but the kurtosis term remains.

Example. (Skewed errors and a misspecified line)

Continue [Example \(Residual sum of squares under a wrong mean\)](#), now with independent centred exponential errors, for which $\sigma^2 = 1$, $\mu_3 = 2$ and $\mu_4 = 9$. [Theorem 2.3.3](#) gives $\text{Var}(\text{RSS}) = 29.80$. Of this, the kurtosis term contributes 16.73 and the skewness term -4.27 . Normal errors with the same mean and variance would give 17.33, so the exponential errors multiply the variance by 1.72 even though the mean of RSS is unchanged. Over 10^6 simulated data sets, the variance of RSS is 30.04. The script also confirms the formula exactly for a two-point error distribution.

```
se = Q.std() / np.sqrt(N)
vals, probs = np.array([-1.0, 2.0]), np.array([2 / 3, 1 / 3])
exact = 0.0
for idx in itertools.product(range(2), repeat=3):
    u = vals[list(idx)]
    y = mu + B @ u
    exact += np.prod(probs[list(idx)]) * (y @ A @ y)
x = np.arange(1.0, 7.0) # n = 6 design points
X = np.column_stack([np.ones_like(x), x])
M = X @ np.linalg.solve(X.T @ X, X.T)
A2 = np.eye(6) - M # RSS = Y'(I - M)Y for a straight
    line
theta = 0.25 * (x - 3.5) ** 2 # true mean is curved: the line is
    wrong
a = np.diag(A2)

def var_quadform(A, theta, m2, m3, m4):
    """Var(Y'AY) for independent Y_i with means theta_i and common central moments.
    """
    a = np.diag(A)
    return ((m4 - 3 * m2**2) * a @ a + 2 * m2**2 * np.trace(A @ A)
            + 4 * m2 * theta @ A @ A @ theta + 4 * m3 * theta @ A @ a)

# A2 = I - M for a straight line at x = 1..6; theta = true (curved) mean
mean_rss = np.trace(A2) + theta @ A2 @ theta # sigma^2 = 1
var_expo = var_quadform(A2, theta, 1.0, 2.0, 9.0) # centred exponential errors
var_norm = var_quadform(A2, theta, 1.0, 0.0, 3.0) # what normal errors would give

E = rng.exponential(size=(N, 6)) - 1.0
R = np.einsum("ij,jk,ik->i", theta + E, A2, theta + E)
print(f"mean {mean_rss:.3f} vs {R.mean():.3f}")
print(f"var {var_expo:.3f} vs {R.var():.3f} (normal-theory value {var_norm:.3f})")
```

[Run this code online](#)

2.3.3 Exercises

2.3.3.1 A. Check your understanding

Exercise A1.

For random vectors $\mathbf{U}$ and $\mathbf{V}$ of the same size, show that $E\|\mathbf{U} - \mathbf{V}\|^2 = \text{tr Cov}(\mathbf{U}) + \text{tr Cov}(\mathbf{V}) - 2 \text{tr Cov}(\mathbf{U}, \mathbf{V}) + \|E \mathbf{U} - E \mathbf{V}\|^2$.

2.3.3.2 B. Practice

Exercise B1.

Let $E(Y_t) = \mu$ and $\text{Cov}(Y_t, Y_s) = \sigma^2 \phi^{|t-s|}$ for $t, s = 1, \dots, n$, with $|\phi| < 1$. Show that the sample variance satisfies

$$E(S^2) = \sigma^2 \left[1 - \frac{2}{n(n-1)} \sum_{k=1}^{n-1} (n-k) \phi^k \right].$$

Is S^2 biased up or down when $\phi > 0$? What happens as $n \rightarrow \infty$ with ϕ fixed?

Solution. With $\mathbf{C} = \mathbf{I} - n^{-1} \mathbf{1} \mathbf{1}^\top$ and $E(\mathbf{Y}) = \mu \mathbf{1}$, $\mathbf{C} E(\mathbf{Y}) = \mathbf{0}$, so $E(\mathbf{Y}^\top \mathbf{C} \mathbf{Y}) = \text{tr}(\mathbf{C} \boldsymbol{\Sigma}) = \text{tr}(\boldsymbol{\Sigma}) - n^{-1} \mathbf{1}^\top \boldsymbol{\Sigma} \mathbf{1}$. Here $\text{tr}(\boldsymbol{\Sigma}) = n\sigma^2$, and $\mathbf{1}^\top \boldsymbol{\Sigma} \mathbf{1}$ sums all entries. There are n diagonal entries and $2(n-k)$ entries at lag k , so $\mathbf{1}^\top \boldsymbol{\Sigma} \mathbf{1} = \sigma^2 [n + 2 \sum_{k=1}^{n-1} (n-k) \phi^k]$. Dividing $n\sigma^2 - n^{-1} \mathbf{1}^\top \boldsymbol{\Sigma} \mathbf{1}$ by $n-1$ gives the formula. For $\phi > 0$ the sum is positive, so S^2 is biased downwards: positively correlated observations cluster, and the sample looks less variable than the process is. As $n \rightarrow \infty$ the sum is $O(n)$, so the bias is $O(1/n)$ and vanishes.

Exercise B2.

Suppose $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$ and $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{V}$. Show that $E(\text{RSS}) = \sigma^2 \text{tr}((\mathbf{I} - \mathbf{M})\mathbf{V})$. For the equicorrelated $\mathbf{V} = (1 - \rho)\mathbf{I} + \rho \mathbf{1} \mathbf{1}^\top$ and a model with $\mathbf{1} \in \mathcal{C}(\mathbf{X})$, show that $E(\text{RSS})/(n-r) = \sigma^2(1 - \rho)$.

Exercise B3.

Let $Y_1, \dots, Y_n$ be independent and identically distributed with variance σ^2 and finite fourth central moment μ_4 . Use [Theorem 2.3.3](#) to show that $\text{Var}(S^2) = (\mu_4 - 3\sigma^4)/n + 2\sigma^4/(n-1)$. Compare with the normal case.

Solution. Take $\mathbf{A} = \mathbf{C}/(n-1)$ and $\boldsymbol{\theta} = \mu \mathbf{1}$. Then $\mathbf{A}\boldsymbol{\theta} = \mathbf{0}$, so the last two terms of [Theorem 2.3.3](#) vanish. Every diagonal entry of $\mathbf{A}$ is $(1 - 1/n)/(n-1) = 1/n$, so $\mathbf{a}^\top \mathbf{a} = 1/n$, and $\text{tr}(\mathbf{A}^2) = \text{tr}(\mathbf{C})/(n-1)^2 = 1/(n-1)$. This gives the formula. Normal

data have $\mu_4 = 3\sigma^4$ and $\text{Var}(S^2) = 2\sigma^4/(n-1)$. Heavy tails ($\mu_4 > 3\sigma^4$) add a term of the same order $1/n$, so the normal-theory variance is wrong even in large samples.

Exercise B4.

Under the conditions of [Theorem 2.3.3](#) with $\boldsymbol{\theta} = \mathbf{0}$, show that $\text{Cov}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}, \mathbf{Y}^\top \mathbf{B} \mathbf{Y}) = (\mu_4 - 3\sigma^4) \mathbf{a}^\top \mathbf{b} + 2\sigma^4 \text{tr}(\mathbf{A} \mathbf{B})$, where $\mathbf{a}$ and $\mathbf{b}$ are the diagonals of the symmetric matrices $\mathbf{A}$ and $\mathbf{B}$. *Hint:* apply the theorem to $\mathbf{A} + \mathbf{B}$.

Exercise B5.

Let $Y_1, \dots, Y_n$ be independent $N(0, \sigma^2)$. Among symmetric $\mathbf{A}$ with $E(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = \sigma^2$, show that $\text{Var}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y})$ is minimized by $\mathbf{A} = n^{-1} \mathbf{I}$. *Hint:* $\text{tr}(\mathbf{A}^2) \geq (\text{tr } \mathbf{A})^2/n$.